Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

HHS Public Access

Author manuscript

Mayo Clin Proc. Author manuscript; available in PMC 2025 March 01.

Published in final edited form as:

Mayo Clin Proc. 2024 March ; 99(3): 375–386. doi:10.1016/j.mayocp.2023.10.002.

Adrenal Hemorrhage: A Comprehensive Analysis of a Heterogeneous Entity – Etiology, Presentation, Management and Outcomes

Prerna Dogra, MBBS[1,2], **Mrunal Chinthapalli, MBBS**[1], **Rashi Sandooja, MBBS**[1], **Leili Rahimi, MD**[1], **Nicole M Iniguez-Ariza, MD**[1,3,4], **Trenton Foster, MD**[5], **Irina Bancos, MD**[1]

[1]Division of Endocrinology, Diabetes, Metabolism, and Nutrition, Mayo Clinic, Rochester, MN, USA

[2]Division of Endocrinology, Diabetes, and Metabolism, University of Wisconsin School of Medicine and Public Health, Madison, Wisconsin, USA

[3]Department of Endocrinology and Metabolism, Instituto Nacional de Ciencias Médicas y Nutrición Salvador Zubirán, Mexico City, Mexico

[4]Department of Medicine, American British Cowdray Medical Center, Mexico City, Mexico.

[5]Department of Surgery, Mayo Clinic, Rochester, MN, USA

Abstract

Objective: To investigate the etiology, presentation, management and outcomes of patients with adrenal hemorrhage (AH) as sparse data are available.

Patients and Methods: Longitudinal study of adult consecutive patients with radiologically confirmed AH (January 1, 2017 to December 31, 2021).

Reprints and correspondence: Irina Bancos, Division of Endocrinology, Diabetes, Metabolism and Nutrition, Mayo Clinic, Rochester, MN 55905, USA. Bancos.Irina@mayo.edu. Phone number: +1 507-286-1387.

Publisher's Disclaimer: This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Preprint Server Disclosure: This work has not been previously published.

Abstract Presentation: The poster for the manuscript was presented at the national conference – ENDO 2023 held in Chicago, Illinois between June 16–18, 2023

CRediT author statement

1. **Prerna Dogra:** Conceptualization, methodology, data curation, validation, visualization, writing – original draft.
2. **Mrunal Chinthapalli:** Investigation, writing – review and editing.
3. **Rashi Sandooja:** Investigation, writing – review and editing.
4. **Leili Rahimi:** Investigation, writing – review and editing.
5. **Nicole M Iniguez-Ariza:** Investigation, writing – review and editing.
6. **Trenton Foster:** Conceptualization, writing – review and editing.
7. **Irina Bancos:** Conceptualization, methodology, software, formal analysis, resources, validation, data curation, writing – review and editing, visualization, supervision.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Results: Of the 363 patients with AH (median age 62 years (IQR 50–70), 128 (35%) women), 338 (93%) had unilateral AH, and 25 (7%) had bilateral AH. AH was discovered incidentally in 152 (42%) patients and during the evaluation of trauma in 103 (28%), abdominal/back pain in 90 (25%), critical illness in 13 (4%), and symptoms of adrenal insufficiency in 5 (1%) patients. Etiologies included postoperative complications (150, 34%), trauma (107, 30%), coagulopathy (22, 6%), anticoagulant/antiplatelet therapy (39, 12%), adrenal neoplasm (22, 6%), and sepsis (11, 3%). Overall, 165 (46%) patients were hospitalized, and no deaths occurred due to AH. Median baseline AH size was 34 mm (24–40) on the right and 29 mm (22–37) on the left. Among 246 patients with follow-up imaging, AH resolution was complete in 155 (63%) patients and incomplete in 74 (31%) at a median of 15 months (6–31). Patients with bilateral AH were more likely to have underlying coagulopathy (44 *vs.* 3%) and develop primary adrenal insufficiency (72 *vs.* 0%) than those with unilateral AH ($P<.0001$).

Conclusion: AH often presents as an incidental unilateral lesion with normal adrenal function, commonly attributed to postoperative complications or trauma. In contrast, bilateral AH is rare and typically linked to underlying coagulopathy, with primary adrenal insufficiency developing in most.

Keywords

adrenal bleeding; adrenal hematoma; adrenal tumor; primary adrenal insufficiency; adrenal function

INTRODUCTION

Due to the unique vascular anatomy and physiological response to acute stress, adrenal glands are particularly susceptible to bleeding.(1–4) Primary adrenal insufficiency, the most dreaded complication of adrenal hemorrhage, is frequently overlooked due to non-specific clinical presentation, coexisting illness and lack of awareness.(5) Prior studies on adrenal hemorrhage have focused on specific etiologies such as trauma, adrenal neoplasms, COVID-19 infection or interventional procedures,(6–8)or has been limited to certain radiological presentation like bilateral adrenal hemorrhage,(9, 10) or hospital setting.(10, 11) As a result, our understanding of the full spectrum of adrenal hemorrhage remains limited. The most extensive study to date examining a broader range of etiologies and presentations was limited to 141 hospitalized patients, with autopsy findings used to diagnose nearly 50% of cases.(11) Additionally, this study was published over two decades ago and excluded patients with underlying adrenal tumors or adrenal hemorrhage resulting from complications of non-adrenal surgeries.(11) Given the knowledge gaps and the evolving healthcare landscape, the objectives of our current study were to investigate the etiological association, clinical and radiological presentation, and outcomes of patients with adrenal hemorrhage.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

METHODS

Study design, patient selection, data collection

This is a retrospective longitudinal cohort study and included adult patients with adrenal hemorrhage from January 1, 2017, to December 31, 2021. The study received approval from the Mayo Clinic institutional review board and complies with the Declaration of Helsinki.

Research consent waiver was used. Patients were identified electronically from the institutional radiological database by searching for the terms ‘adrenal’ combined with ‘hemorrhage,’ ‘hemorrhagic,’ ‘hematoma,’ and ‘bleeding’ within a 50-word space limit of the entire radiological report. Individuals under the age of 18 were excluded.

Information regarding medical records for various factors, including demographics, clinical presentation, etiological associations, healthcare setting (hospital or outpatient), imaging characteristics (hemorrhage laterality, size, resolution), management strategies (no therapy *vs.* conservative *vs.* procedural or surgical intervention), and outcomes (primary adrenal insufficiency and mortality) was obtained from review of medical records. Adrenal hemorrhage was diagnosed in accordance with the radiology report and there was no blinded secondary review. When available, cross-sectional abdominal imaging preceding and following the hemorrhagic event were reviewed to help differentiate adrenal hemorrhage from neoplasm based on (i) prior imaging showing normal adrenal glands, (ii) a dramatic change in the adrenal neoplasm size/appearance, (iii) subsequent imaging showing resolving or resolved hemorrhage, and/or (iv) evidence of active extravasation. Adrenal hemorrhage occurring in a metachronous manner (unilateral adrenal hemorrhage followed by involvement of the contralateral adrenal gland later), were classified as bilateral adrenal hemorrhage as long as they were diagnosed during the same event (<3 months) and the etiology of the adrenal hemorrhage was the same. In cases where multiple potential etiologies or presentations were present, the one with the highest likelihood was considered following a detailed review of the clinical case scenario. Healthcare setting was classified as ‘hospitalization’ if the discovery of adrenal hemorrhage occurred during or was immediately followed by hospitalization. Conservative management encompassed intravenous fluids, blood transfusion, pain management, or imaging surveillance for adrenal hemorrhage. The size of hemorrhage was determined by measuring the largest diameter on the initial imaging study. Last available imaging study was used to determine the evolution of adrenal hemorrhage (resolution, decrease or increase in size, stability). Subgroup analyses were conducted on hemorrhage laterality (unilateral *vs.* bilateral), clinical presentation (incidental *vs.* non-incidental), healthcare setting at discovery (outpatient *vs.* hospitalization), and in patients with bilateral adrenal hemorrhage, adrenal function (primary adrenal insufficiency *vs.* normal adrenal function).

Statistical analysis

Categorical data were presented as counts and percentages, while continuous data were described using medians and interquartile ranges (Q1-Q3). The association between variables was examined using the chi-square test and the Wilcoxon/Kruskal Wallis test.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

A *p*-value of less than 0.05 was considered statistically significant. All statistical analyses were performed using the JMP Software, version 16.1.0, SAS Institute Inc.

RESULTS

Clinical characteristics

During the 5-year study period, a total of 363 patients (median age 62 years [52–70], men 235 [65%]) were diagnosed with adrenal hemorrhage. Of the total patients, 338 (93%) had unilateral adrenal hemorrhage, while 25 (7%) had bilateral adrenal hemorrhage. Among those with bilateral adrenal hemorrhage, the appearance of the hemorrhage was synchronous in 20 (80%) patients and metachronous in 5 (20%) patients. The most common presentation of adrenal hemorrhage was incidental, observed in 152 (42%) patients. This was followed by trauma evaluation in 103 (28%) patients, abdominal or back pain in 90 (25%), critical illness in 13 (4%), and symptoms of adrenal insufficiency or crisis in 5 (1%) patients. Regarding the primary etiology of adrenal hemorrhage, postoperative or procedural complications were identified in 150 (41%) patients, trauma in 107 (30%), coagulopathic disorder in 22 (6%), anticoagulant and/or antiplatelet therapy in 39 (11%), adrenal neoplasm in 22 (6%), and sepsis in 11 (3%) patients. The etiology remained unclear (idiopathic) in 12 (3%) patients. Liver transplantation (n=65) was the most common surgery, antiphospholipid syndrome (n=13) was the most common coagulopathic disorder, motor vehicle accidents (n=58) were the most common cause of trauma, and metastasis (n=15) was the most common adrenal neoplasm associated with adrenal hemorrhage. No difference in etiology was seen between men and women (p=.54). Of the total patients, 165 (46%) patients were hospitalized. The majority of patients were managed conservatively (260, 71%) or required no therapy (93, 26%), but in a small proportion (10, 3%), active extravasation necessitated procedural or operative intervention (embolization in 6, unilateral adrenalectomy in 1, stent placement in 1, explorative laparotomy in 2).

Patients with bilateral adrenal hemorrhage were more likely to have underlying coagulopathy (44 *vs.* 3%) or sepsis (12 *vs.* 2%) and present with abdominal or back pain (48 *vs.* 23%), critical illness (12 *vs.* 3%), or symptoms of adrenal insufficiency or crisis (12 *vs.* 0%) than those with unilateral adrenal hemorrhage (p<.0001); however, hospitalization rates (56 *vs.* 45%, p=.30) were similar between the two groups. Incidental presentation was more common in unilateral adrenal hemorrhage (44 *vs.* 8%, p<.0001) and postoperative or procedural complications (72 *vs.* 17%, p<.0001). Patients with nonincidental presentation (70 *vs.* 14%, p<.0001) and especially those with traumatic etiology (59 *vs.* 5%, p<.0001) were more likely to be hospitalized. (Table 1)

Radiological characteristics

In patients with unilateral adrenal hemorrhage, there was a propensity for hemorrhage to occur more commonly in the right adrenal gland (n=218) than the left adrenal gland (n=120). When considering the entire cohort, the baseline median hemorrhage size on the right (n=248) was 34 mm (24–40), and on the left (n= 145) was 29 mm (22–37). In the 246 patients with follow-up imaging, the hemorrhage size decreased by a median of −8.6 mm (−21.8 to −3.2) per 6 months on the right and by a median of −6.9 mm (−15 to −2.8) on the

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

left. At a median imaging follow-up of 15 months (16–31), complete hemorrhage resolution was seen in 156 (63%) and incomplete in 74 (31%) patients, whereas in 11 (4%) there was no change in size, and 6 (2%) had an increase in size.

Although the median baseline hemorrhage size in patients with bilateral adrenal involvement was larger on both the right (40 *vs.* 32 mm, p=.01) and left side (35 *vs.* 27 mm, p=.03) than in unilateral adrenal hemorrhage, the change in size per 6 months was not different. Considering the larger baseline size, complete hemorrhage resolution on the last follow-up imaging was lower (50 *vs.* 64%, p<.001) in patients with bilateral adrenal hemorrhage. No difference was seen in size change per 6 months based on presentation or the healthcare setting.

Outcomes

Of the 25 patients with bilateral adrenal hemorrhage, 18 (72%) developed primary adrenal insufficiency. The diagnosis of adrenal insufficiency was based on low serum cortisol and elevated corticotropin (ACTH) in 11 patients, abnormal cosyntropin (synthetic ACTH) stimulation test in 2, clinical suspicion and low cortisol in 2, and clinical suspicion alone in 2 patients. Patients with bilateral adrenal hemorrhage who developed primary adrenal insufficiency were more likely to have an underlying coagulopathy (61 *vs.* 0%, p=.02) than those with normal adrenal function. Both groups were otherwise similar in demographics, presentation (incidental *vs.* nonincidental), and imaging characteristics, including radiographic resolution of hemorrhage. (Table 2) Among the 18 patients with bilateral adrenal hemorrhage who developed primary adrenal insufficiency, only one patient had documented recovery of adrenal function at 11 months, as evidenced by the successful weaning off of glucocorticoid replacement therapy. However, it is important to note that not all patients attempted weaning or were re-tested for recovery.

None of the 338 patients with unilateral adrenal hemorrhage required initiation of glucocorticoid replacement therapy (surrogate for diagnosis of primary adrenal insufficiency) at diagnosis or follow-up. Of note, only 18 (5%) of the 338 patients had laboratory evaluation with measurement of serum cortisol and/or ACTH and these results were within the parameters of normal adrenal function.

Overall mortality was 64 (18%) in the entire cohort, but none were due to adrenal hemorrhage and occurred at a median of 10 months (2–19) following the onset of adrenal hemorrhage. Although patients with bilateral adrenal hemorrhage had a higher mortality rate (28 *vs.* 17%, p=.17) and a shorter median time to death (6 *vs.* 11 months, p=.38) compared to those with unilateral hemorrhage, the difference was not statistically significant.

DISCUSSION

In this largest-to-date longitudinal study on adrenal hemorrhage, we report on the clinical presentation, etiological associations, imaging characteristics, management, and outcomes in a consecutive cohort of adult patients. We found that most adrenal hemorrhages were unilateral, presenting incidentally or during trauma evaluation, and occurred secondary to postoperative complication or trauma. In the 7% of patients with bilateral hemorrhage,

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

abdominal or back pain was the most common presentation, and underlying coagulopathy was the most common etiology. Primary adrenal insufficiency developed in three-fourths of patients with bilateral adrenal hemorrhage and recovery of adrenal function was rare. Interventional or operative treatment was rarely required, and most patients were managed conservatively. Overall mortality was high, but none were due to adrenal hemorrhage.

Our study shows that multiple factors can precipitate adrenal hemorrhage, which occurs more often as a consequence rather than as the primary inciting event. The three suprarenal arteries supplying each adrenal gland form a rich vascular plexus on entering the adrenal cortex and, via relatively fewer corticomedullary venous channels, drain into the medullary sinusoids, which ultimately empty into a single central medullary vein (Figure 1–2). This vascular layout creates a dam that can easily become overloaded by increased perfusion pressure or venoconstriction. During periods of physiological stress, it is possible that corticotropin and catecholamine release may have a role in vascular congestion.(2–4) Corticotropin increases the blood supply to adrenal glands, stimulating cortisol production, and the release of catecholamine into the medullary vein results in venoconstriction and increased back venous pressure. This notable interconnection of the vasculature and physiology potentially predisposes adrenal glands to vascular congestion, hemorrhage, and necrosis.

In the 25-year case series of 141 patients with adrenal hemorrhage reported by Vella *et al.* from our institution more than two decades ago, fewer than 25% of patients presented incidentally on imaging or following trauma, and over 50% had bilateral hemorrhage.(11) The striking differences in the presentation and hemorrhage laterality between the two studies from the same institution over a span of a half-century underscore the changing landscape of healthcare, guided primarily by the increasing use of radiological investigations in recent decades.(12) This also explains the relatively older age of our cohort (between 50 and 70 years), as comorbidities and illnesses prompt more frequent use of imaging. Even though Vella *et al.* reported only on hospitalized patients, we found no difference in the prevalence of unilateral vs. bilateral adrenal hemorrhage based on the healthcare setting in our cohort.

Postoperative complications (liver transplantation and hepatectomy) and trauma (motor vehicle accidents and falls) were the most common etiologies in our cohort. In concordance with other studies on adrenal hemorrhage due to trauma and liver transplantation, we found right adrenal hemorrhage to be twice as common as left adrenal hemorrhage.(13–15) The right suprarenal vein is short and drains directly into the inferior vena cava, and as such, increased pressure, obstruction, or thrombosis in the inferior vena cava would prevent drainage from the right adrenal gland resulting in venous congestion and hemorrhagic necrosis. By comparison, the left suprarenal vein is longer and drains into the left renal vein before combining with inferior vena cava.(14) (Figure 1) Furthermore, the anatomical location of the right adrenal gland between the liver and lower thoracic vertebrae makes it more prone to compression.(14) Orthotopic liver transplantation typically involves ligation and division of the right suprarenal vein that could result in right adrenal infarction or hemorrhage.(15) In our study, men had a higher propensity to develop adrenal hemorrhage, specifically unilateral adrenal hemorrhage. The exact reason for this is not entirely apparent,

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

and unlike other studies on traumatic adrenal hemorrhage, we did not find trauma to be a factor for the observed male-to-female distribution.(7, 13)

Adrenal neoplasms were an uncommon cause of adrenal hemorrhage in our cohort, with the majority attributed to underlying adrenal metastases and rarely (<5%) pheochromocytoma. Our findings differ from Marti *et al.*, who reviewed 139 patients with hemorrhagic adrenal neoplasms, including six cases from their center and 133 cases from published literature, and found that pheochromocytomas represented almost half of the neoplasms, and only 14% were adrenal metastases.(8) As most of the data in the study by Marti *et al.* are from a systematic review of case reports, there is a high likelihood of publication bias related to exceptional cases typically reported in the literature.(8) It has been generally recommended to obtain interval imaging to evaluate for an underlying adrenal mass that may not be visible at the time of hemorrhage occurrence.(5) In our study, we found adrenal hemorrhage decreased in size at a rate of 6–9 mm per 6 months, with the majority having entirely resolved in a little over a year. (Figure 3) To the best of our knowledge, this is the first study reporting on the rate and timeline of hemorrhage resolution.

Similar to Vella *et al.*, we found a high prevalence of primary adrenal insufficiency (72%) in patients with bilateral adrenal hemorrhage.(11) The fact that one-third of the adrenocortical tissue is enough to maintain normal adrenal function explains why no patient with unilateral and one-fourth with bilateral adrenal hemorrhage did not develop primary adrenal insufficiency.(16) The most common etiology of bilateral adrenal hemorrhage was an underlying coagulopathy, with antiphospholipid syndrome being the most common, and was the only factor associated with development of primary adrenal insufficiency. Notably, anticoagulation or antiplatelet therapy was not associated with hemorrhage laterality and represented a relatively less common cause of adrenal hemorrhage overall. During a median follow-up of two years, 18% of patients died, but none because of adrenal hemorrhage. This relatively high mortality hints at this population's overall poor health status (due to underlying illness), especially those with bilateral hemorrhage. While the higher mortality rate and shorter time to death in patients with bilateral hemorrhage were not statistically different from those with unilateral hemorrhage, this could be related to small sample size.

In addition to being the largest cohort on adrenal hemorrhage, the notable strengths of our study are the inclusion of consecutive patients irrespective of etiology, presentation, hemorrhage laterality, and healthcare setting. The broad inclusion criteria and longitudinal follow-up helped provide an in-depth view of the various clinical and radiological aspects of adrenal hemorrhage, making our study unique. Our study has limitations, the most obvious of which are the retrospective single-center study design, referral bias, and the predominantly White population that could limit the generalizability of our findings. Given that we identified patients with adrenal hemorrhage using our radiological database, the prevalence of hemorrhagic adrenal neoplasm may be underestimated in our cohort, as the same is not always evident on imaging. We also acknowledge the lack of uniform criteria for the diagnosis of primary adrenal insufficiency, and that a significant proportion of patients did not undergo necessary laboratory evaluation to assess adrenal function. Nevertheless, among those who received the diagnosis of primary adrenal insufficiency, the majority had an irrefutable diagnosis supported by laboratory evidence.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

In conclusion, adrenal hemorrhage is a heterogeneous condition, predominantly presenting as an incidental unilateral lesion with normal adrenal function and occurring mainly as a postoperative complication or following trauma. By comparison, bilateral adrenal hemorrhage is rare, commonly manifesting as abdominal or back pain, and resulting from underlying coagulopathy, with primary adrenal insufficiency developing in the majority of patients. Thus, initiation of glucocorticoid replacement should be considered in all patients with bilateral adrenal hemorrhage until normal endogenous cortisol production can be confirmed. In general, adrenal hemorrhage is a self-limiting process effectively managed conservatively, but surgical or procedural intervention may be required in rare instances of active extravasation and hemodynamic instability.

Funding:

This research was partly supported by the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) and National Institute of Aging (NIA) of the National Institutes of Health (NIH) USA under award K23DK121888, R03DK132121, and R03AG71934 (to I.B). The views expressed are those of the author(s) and not necessarily those of the National Institutes of Health USA.

Declaration of interest:

IB reports consulting (fee to institution) from HRA Pharma, Recordati, Corcept, Sparrow, Xeris, Astra Zeneca, Neurocrine, Diurnal, Adrenas, Spruce, outside this work. IB reports funding for investigator-initiated award from Recordati (outside this work). Other authors declare no conflict of interest.

ABBREVIATIONS:

ACTH	corticotropin

REFERENCES

1. Dobbie JW, Symington T. The human adrenal gland with special reference to the vasculature. J Endocrinol. 1966;34(4):479–89. [PubMed: 5933099]
2. Wilbur OM Jr., Rich AR. A study of the role of adrenocorticotropic hormone (ACTH) in the pathogenesis of tubular degeneration of the adrenals. Bull Johns Hopkins Hosp. 1953;93(5):321–47. [PubMed: 13106585]
3. Szabo S, McComb DJ, Kovacs K, Huttner I. Adrenocortical hemorrhagic necrosis: the role of catecholamines and retrograde medullary-cell embolism. Arch Pathol Lab Med. 1981;105(10):536–9. [PubMed: 6895166]
4. Haller RM. Experimental adrenocortical hemorrhage in anticoagulated rats induced by stimulating the adrenal cortex [in German]. Virchows Arch. Abt. A Path. Anat 1969;346:204–11.
5. Dogra P, Sundin A, Juhlin CC, Calissendorff J, Falhammar H, Bancos I. Rare benign adrenal lesions. Eur J Endocrinol. 2023.
6. Elhassan YS, Iqbal F, Arlt W, Baldeweg SE, Levy M, Stewart PM, et al. COVID-19-related adrenal haemorrhage: Multicentre UK experience and systematic review of the literature. Clin Endocrinol (Oxf). 2023.
7. Falhammar H, Koskinen SK, Kistner A. Adrenal trauma experience at a major tertiary centre in Sweden: Clinical and radiological findings. Clin Endocrinol (Oxf). 2022;97(1):28–35. [PubMed: 35174528]
8. Marti JL, Millet J, Sosa JA, Roman SA, Carling T, Udelsman R. Spontaneous adrenal hemorrhage with associated masses: etiology and management in 6 cases and a review of 133 reported cases. World J Surg. 2012;36(1):75–82. [PubMed: 22057755]
9. Rao RH, Vagnucci AH, Amico JA. Bilateral massive adrenal hemorrhage: early recognition and treatment. Ann Intern Med. 1989;110(3):227–35. [PubMed: 2643380]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

10. Kovacs KA, Lam YM, Pater JL. Bilateral massive adrenal hemorrhage. Assessment of putative risk factors by the case-control method. Medicine (Baltimore). 2001;80(1):45–53. [PubMed: 11204502]
11. Vella A, Nippoldt TB, Morris JC 3rd. Adrenal hemorrhage: a 25-year experience at the Mayo Clinic. Mayo Clin Proc. 2001;76(2):161–8. [PubMed: 11213304]
12. Ebbehoj A, Li D, Kaur RJ, Zhang C, Singh S, Li T, et al. Epidemiology of adrenal tumours in Olmsted County, Minnesota, USA: a population-based cohort study. Lancet Diabetes Endocrinol. 2020;8(11):894–902. [PubMed: 33065059]
13. Mehrazin R, Derweesh IH, Kincade MC, Thomas AC, Gold R, Wake RW. Adrenal trauma: Elvis Presley Memorial Trauma Center experience. Urology. 2007;70(5):851–5. [PubMed: 18068438]
14. Sinelnikov AO, Abujudeh HH, Chan D, Novelline RA. CT manifestations of adrenal trauma: experience with 73 cases. Emerg Radiol. 2007;13(6):313–8. [PubMed: 17252249]
15. Bowen AD, Keslar PJ, Newman B, Hashida Y. Adrenal hemorrhage after liver transplantation. Radiology. 1990;176(1):85–8. [PubMed: 2191375]
16. Brauckhoff M, Gimm O, Thanh PN, Bar A, Ukkat J, Brauckhoff K, et al. Critical size of residual adrenal tissue and recovery from impaired early postoperative adrenocortical function after subtotal bilateral adrenalectomy. Surgery. 2003;134(6):1020–7; discussion 7–8. [PubMed: 14668736]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

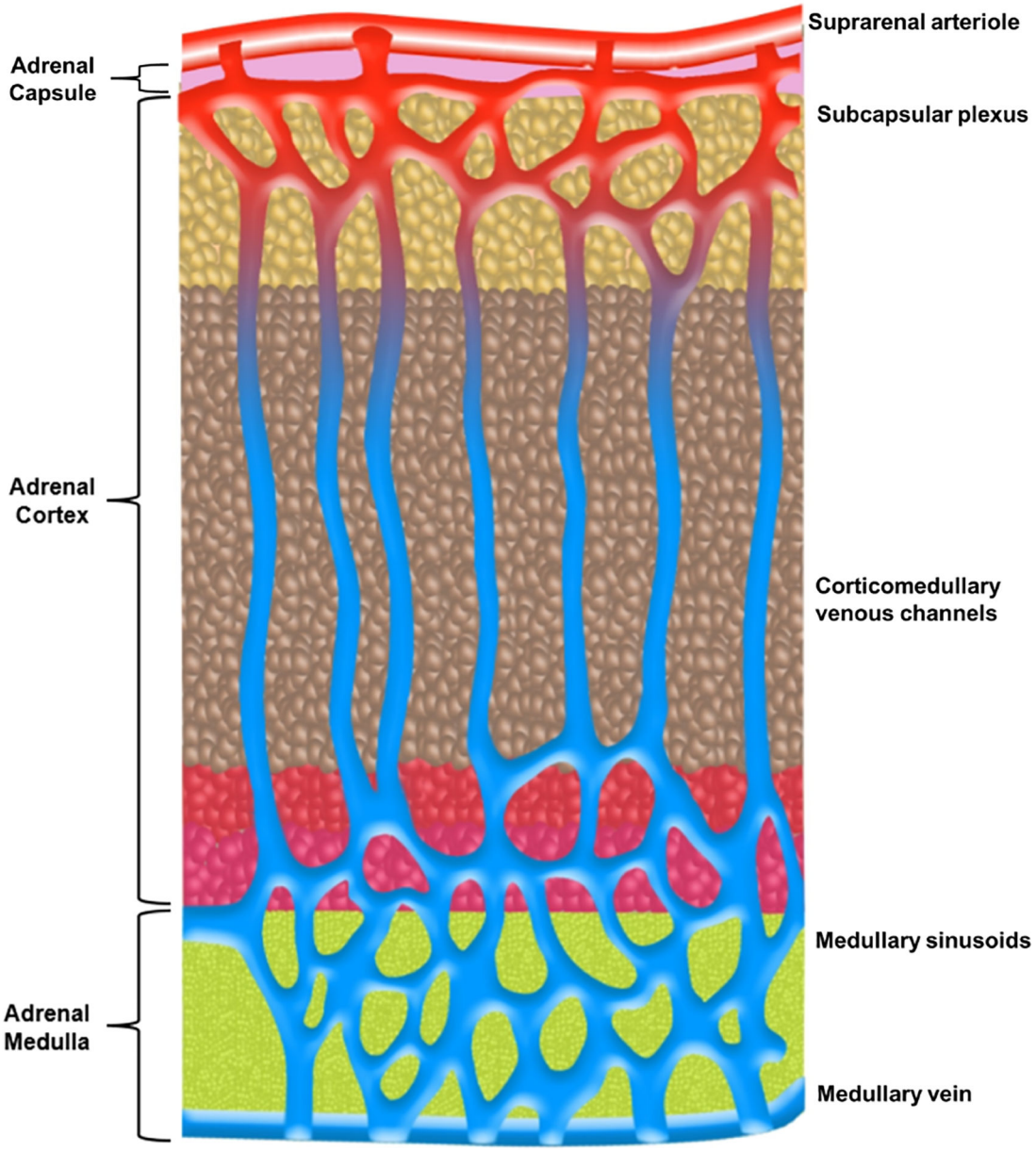


Figure 1:
Adrenal gland vascular anatomy

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

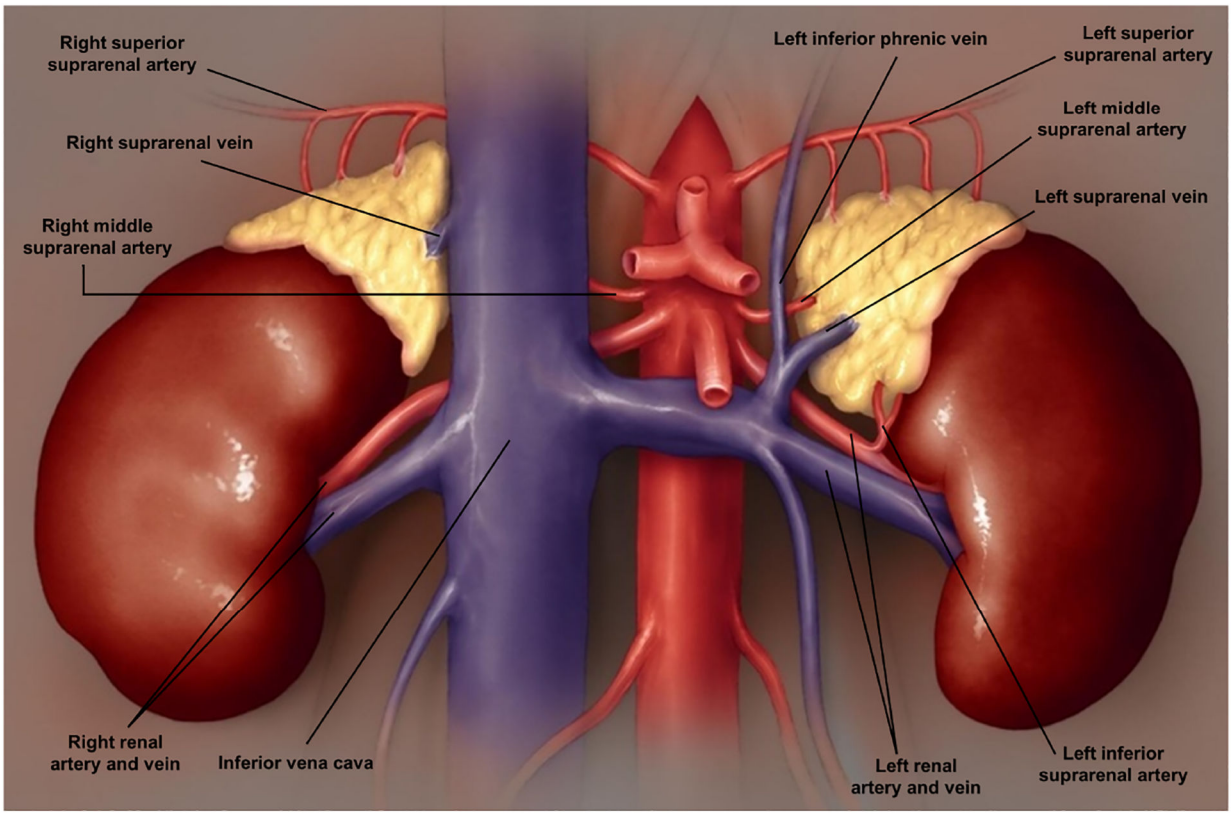


Figure 2:
Illustration of intra-adrenal vasculature (‘vascular dam’)

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

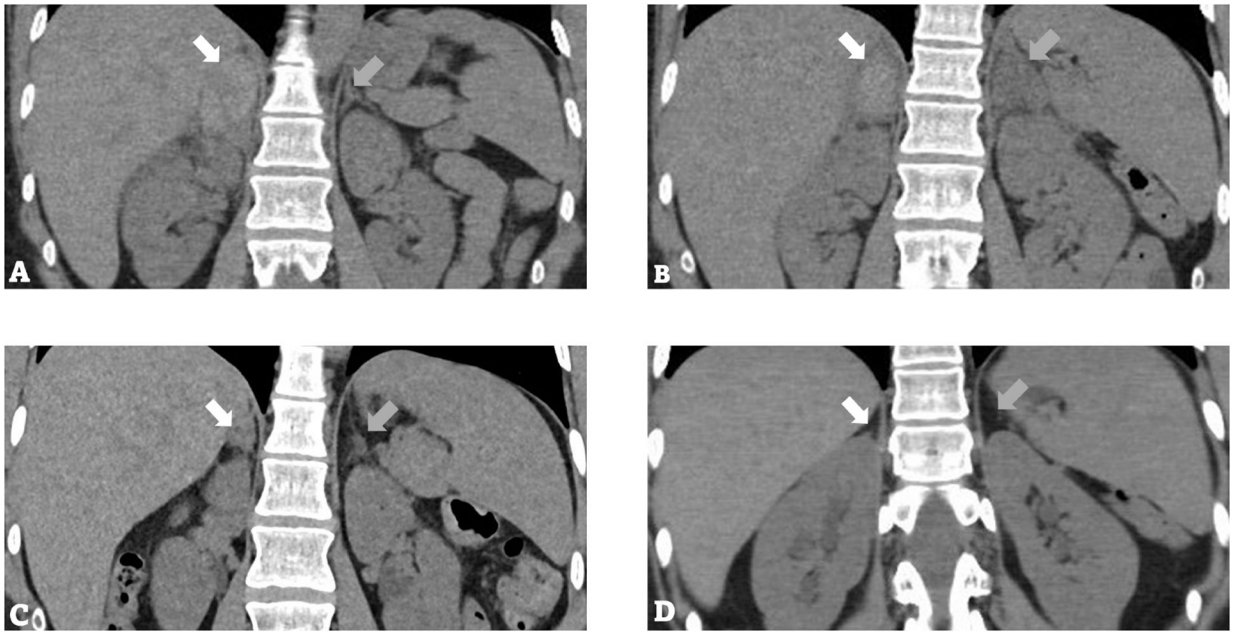


Figure 3:

Evolution of metachronous bilateral adrenal hemorrhage in a 47-year-old patient who developed primary adrenal insufficiency.

Footnote: Appearance and resolution of bilateral adrenal hemorrhage on unenhanced computed tomography (right adrenal – white arrow; left adrenal – grey arrow). (A) 5.2 cm right adrenal hematoma. Note the normal left adrenal gland. (B) Sequential appearance of 4.8 cm left adrenal hematoma (after 2-weeks). (C) Incomplete resolution of bilateral adrenal hemorrhage (after 3-months). (D) Complete resolution of bilateral adrenal hemorrhage (after 11-months). Adrenal glands are atrophic and barely visible.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Mayo Clin Proc. Author manuscript; available in PMC 2025 March 01.

Table 1.

Clinical and radiological characteristics of adrenal hemorrhage overall and based on laterality, presentation, and healthcare setting

	Total N=363	Laterality			Presentation			Healthcare Setting		
		Unilateral N=338	Bilateral N=25	*P*-value	Incidental N=160	Nonincidental N=203	*P*-value	Not hospitalized N=198	Hospitalized[a] N=165	*P*-value
Clinical characteristics										
Median age at diagnosis, years (IQR)	62 (52–70)	62 (51–70)	59 (53–67)	.57	64 (55–71)	61 (48–69)	.03	63 (54–70)	60 (47–70)	.03
Sex, n (%)										
Male	235 (65)	222 (66)	13 (48)	.19	105 (66)	130 (64)	.10	125 (63)	110 (67)	.48
Female	128 (35)	116 (34)	12 (52)		55 (34)	73 (36)		73 (37)	55 (34)	
Race, n (%)										
White	334 (92)	312 (92)	22 (88)		153 (96)	181 (89)		189 (96)	145 (88)	
Black	3 (1)	4 (1)	1 (4)	.15	1 (<1)	2 (1)	.25	1 (<1)	2 (1)	.07
Asian	4 (1)	2 (1)	0 (0)		1 (<1)	3 (2)		1 (<1)	3 (2)	
Other or mixed	22 (6)	20 (6)	2 (8)		5 (3)	17 (8)		6 (3)	14 (9)	
Presentation, n (%)										
Incidental	152 (42)	149 (44)	3 (8)					136 (69)	16 (10)	
Trauma evaluation	103 (28)	101 (30)	2 (8)					11 (6)	92 (56)	
Abdominal/back pain	90 (25)	78 (23)	12 (48)	<.0001	-	-	-	48 (24)	42 (26)	<.0001
Critical illness	13 (4)	10 (3)	3 (12)					1 (<1)	12 (7)	
Symptoms of adrenal insufficiency/crisis	5 (1)	0 (0)	3 (12)					2 (1)	3 (2)	
Hospitalized[a], n (%)	165 (46)	151 (45)	14 (56)	.30	23 (14)	142 (70)	<.0001	-	-	-
Etiology, n (%)										
Postoperative or procedural[b]	150 (41)	147 (44)	3 (12)		115 (72)	35 (17)		125 (63)	25 (15)	
Trauma[c]	107 (30)	105 (31)	2 (8)	<.0001	2 (1)	105 (52)	<.0001	10 (5)	97 (59)	<.0001
Coagulopathic disorder[d]	22 (6)	11 (3)	11 (44)		8 (5)	14 (7)		16 (8)	6 (4)	
Anticoagulation and/or antiplatelet therapy[e]	39 (11)	36 (11)	3 (12)		21 (13)	18 (9)		25 (13)	14 (9)	

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

	Total N=363	Laterality			Presentation			Healthcare Setting		
		Unilateral N=338	Bilateral N=25	*P*-value	Incidental N=160	Nonincidental N=203	*P*-value	Not hospitalized N=198	Hospitalized[a] N=165	*P*-value
Adrenal neoplasm[f]	22 (6)	19 (6)	3 (12)		9 (6)	13 (6)		13 (7)	9 (6)	
Sepsis	11 (3)	8 (2)	3 (12)		1 (<1)	10 (5)		2 (1)	9 (6)	
Idiopathic	12 (3)	12 (4)	0 (0)		4 (3)	8 (4)		7 (4)	5 (3)	
Management, n (%)										
Conservative	260 (71)	329 (70)	22 (88)		98 (61)	162 (80)		121 (61)	139 (84)	
Embolization	6 (2)	6 (2)	0 (0)		0	6 (3)		0	6 (4)	
Adrenalectomy	1 (1)	1 (<1)	0 (0)	.31	0	1 (<1)	<.0001	0	1 (<1)	<.0001
Other[g]	3 (<1)	3 (<1)	0 (0)		2 (1)	1 (<1)		1 (<1)	2 (1)	
None	93 (26)	90 (27)	3 (12)		60 (38)	33 (16)		76 (39)	17 (10)	
Median clinical follow-up, months (IQR), n=265	5 (1–13)	4 (1–12)	7 (3–27)	.06	6 (3–13)	3 (1–12)	.05	6 (3–13)	3 (1–12)	.02
Radiological characteristics										
Laterality, n (%)										
Right	218 (60)	-	-	-	96 (60)	122 (60)	.005	110 (56)	108 (66)	.03
Left	120 (33)				60 (38)	60 (30)		77 (39)	43 (26)	
Bilateral	25 (7)				4 (3)	21 (10)		11 (6)	14 (9)	
Median hemorrhage size at baseline, mm (IQR)										
Right, n=242	34 (24–40)	32 (23–40)	40 (35–45)	.01	30 (21–40)	35 (25–42)	.03	30 (22–40)	35 (25–41)	.21
Left, n=145	29 (22–37)	27 (21–36)	35 (26–40)	.03	27 (21–37)	30 (23–38)	.32	27 (21–36)	31 (25–44)	.02
Follow-up imaging available, n (%)	246 (68)	224 (66)	22 (88)	.02	129 (80)	117 (58)	<.0001	155 (78)	91 (55)	<.0001
Median imaging follow-up duration, months (IQR)	15 (6–31)	15 (7–31)	9 (5–35)	.53	19 (10–33)	9 (4–27)	.003	17 (9–33)	8 (2–26)	.0002
Median hemorrhage size change (in mm) per 6-months, (IQR)										
Right, n=166	−8.6 (−21.8 to −3.2)	−8.6 (−42.3 to −3.1)	−9.3 (−20 to −3.3)	.90	−8.1 (−19.2 to −3.4)	−10.6 (−25 to −3.1)	.58	−8.2 (−19.7 to −3.5)	−10.4 (−25 to −3.1)	.86
Left, n=104	−6.9 (−15 to −2.8)	−7.0 (−15.1 to −3.2)	−6.0 (−15 to −2.6)	.60	−5.8 (−11.4 to −2.3)	−8.2 (−36.4 to −3.2)	.13	−5.6 (−12 to −2.4)	−8.7 (−53.2 to −4.1)	.07

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

	Total N=363	Laterality			Presentation			Healthcare Setting		
		Unilateral N=338	Bilateral N=25	*P*-value	Incidental N=160	Nonincidental N=203	*P*-value	Not hospitalized N=198	Hospitalized[a] N=165	*P*-value
Hemorrhage resolution at last available imaging, n (%)										
Complete resolution	155 (63)	144 (64)	11 (50)	<.0001	85 (65)	70 (60)	.12	106 (68)	52 (34)	.05
Incomplete resolution	74 (31)	67 (30)	8 (36)		38 (30)	36 (31)		44 (28)	30 (33)	
No change in size	11 (4)	11 (5)	0 (0)		5 (4)	6 (4)		4 (3)	7 (8)	
Increased size	6 (2)	3 (1)	3 (14)		1 (1)	5 (4)		1 (1)	5 (6)	

[a]Discovery of adrenal hemorrhage during or followed by hospitalization.

[b]Liver transplant (n=65), hepatectomy (n=17), nephrectomy (n=29), pancreatectomy and splenectomy (n=8), pancreatectomy (n=3), splenectomy (n=3), cholecystectomy (n=3), renal artery stenting (n=1), infrarenal endarterectomy (n=1), other surgeries (n=20)

[c]Motor vehicle accident (n=58), traumatic fall (n=35), sports-related (n=3), other (n=10)

[d]Antiphospholipid syndrome (n=13), thrombocytopenia (n=5), liver disease (n=2), disseminated intravascular anticoagulation (n=1), unspecified coagulopathy (n=1)

[e]Intravenous heparin (n=6), subcutaneous heparin (n=11), warfarin (n=11), direct oral anticoagulant (n=6), aspirin (20), prasugrel (n=1). Note – Patient could be on different anticoagulants or on a combination of anticoagulant and antiplatelet therapy.

[f]Adrenal metastasis (n=15), adenoma (n=3), myelolipoma (n=3), benign adrenal cyst (n=1), pheochromocytoma (n=1)

[g]Explorative laparotomy (n=2), adrenal stent placement (n=1)

Mayo Clin Proc. Author manuscript; available in PMC 2025 March 01.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Table 2:

Patients with bilateral adrenal hemorrhage (n=25) with and without primary adrenal insufficiency

	Primary adrenal insufficiency	Normal adrenal function	*P*-value
	18 (72%)	7 (28%)	
Median age at diagnosis, years (IQR)	59 (52–65)	66 (53–72)	.32
Male sex, n (%)	9 (50)	4 (57)	.10
White race, n (%)	15 (83)	7 (100)	.55
Presentation, n (%)			
Incidental	2 (11)	2 (29)	.30
Nonincidental	16 (89)	5 (71)	
Hemorrhage sequence, n (%)			
Synchronous	14 (78)	6 (86)	.65
Metachronous	4 (22)	1 (14)	
Etiology, n (%)			
Postoperative or procedural	1 (6)	2 (29)	
Trauma	1 (6)	1 (14)	
Coagulopathic disorder	11 (61)	0 (0)	.02
Anticoagulation and/or antiplatelet therapy	1 (6)	2 (29)	
Adrenal neoplasm	2 (11)	1 (14)	
Sepsis	2 (11)	1 (14)	
Median clinical follow-up duration, months (IQR), n=21	8 (6–43)	3 (0.5–16)	.06
Median hemorrhage size at initial discovery, mm (IQR)			
Right	40 (38–44)	40 (34–47)	.86
Left	35 (23–40)	35 (26–48)	.54
Median imaging follow-up duration, months (IQR), n=22	13 (3–57)	9 (5–19)	.83
Median hemorrhage size change (in mm) per 6-months, (IQR), n=22			
Right	−6.2 (−12.5 to −2.4)	−20.4 (−29.6 to −7.7)	.07
Left	−5.7 (−12.4 to −2.4)	−9.4 (−23.4 to −2.6)	.58
Hemorrhage resolution at last available imaging, n (%)			
Complete resolution	8 (50)	3 (50)	
Incomplete resolution	5 (31)	3 (50)	.34
No change in size	0 (0)	0 (0)	
Increased size	3 (19)	0 (0)	
Recovery from primary adrenal insufficiency, n (%)	1 (6%) at 11 months follow up	-	-